ME 101 Introduction to Mechanical Engineering

Engineering Design

Lecture Slides #2

Class Discussion

Weekly Schedule

What's a University?		22 Eylül 2026
Engineering Profession	*Lecture 1*	*29 Eylül 2026*
Engineering Design	Lecture 2	06 Ekim 2026
Manufacturing	Lecture 3	13 Ekim 2026
Effective Writing Skills and Presentation	Lecture 4	20 Ekim 2026
Mechanics	Lecture 5	27 Ekim 2026
Project Planning Team Work	Lecture 6	03 Kasım 2026
Vize		07 - 15 Kasım 2026
Strength,Stress and Materials	Lecture 7	17 Kasım 2026
Practice		24 Kasım 2026
Ethics	Lecture 8	01 Aralık 2026
Project Presentation		08 Aralık 2026
Problem Solving Methods	Lecture 9	15 Aralık 2026
Dynamics, Mechanisms and Control	Lecture 10	22 Aralık 2026
Thermofluids and Energy	Lecture 11	29 Aralık 2026
Final		04 - 17 Ocak 2027
Bütünleme		19 - 24 Ocak 2027

What is a Product?

- Solution for the needs of the society produced by the design process.
- For example; Consumer products, Industry products, Medical products, Transportation products, Defense related products



www.tesla.com



Baxter Dialysis



Self Stirring Mug

What Drives Product Design?

- There is a continuous need for new, cost-effective, high-quality products.
- New products are critical for corporate success.
  - E.g. Samsung released 56 models of smartphones in 2014!



http://www.gsmarena.com/results.php3?nYearMin=2014&nYearMax=2014&sMakers=9

- Successful new products are profitable.
- Some major drivers of product innovation
  - Technological advances
  - Changing customer needs

A Formal Definition of Design

- A new product is the result of a design activity.
- "Design establishes and defines solutions to and applicable structures for problems not solved before, or new solutions to problems which have previously been solved in a different way."
  J.F.Blumrich, Science, 1970

- Engineering design involves
  - An identified need or problem
  - Engineering activities such as analysis, modeling, and testing
  - Providing a solution, generally in form of a product

- But what is a "good" design?

Good Design: Lamborghini



Lamborghini Car Design and Production

The Life of a Bolt

"Bad" Designs

Fukushima Nuclear Reactor



Design Tombola: What is Good Design?

- Take a couple of minutes and list general characteristics of a "good" design
- What we are looking for are *specific well-defined adjectives:*
  - *Example: Functional*
- You will submit your list to a shared pool. Follow the instructions in class.

“Bad” Designs







"Bad" Designs





Goldfish Walker





Privacy Scarf

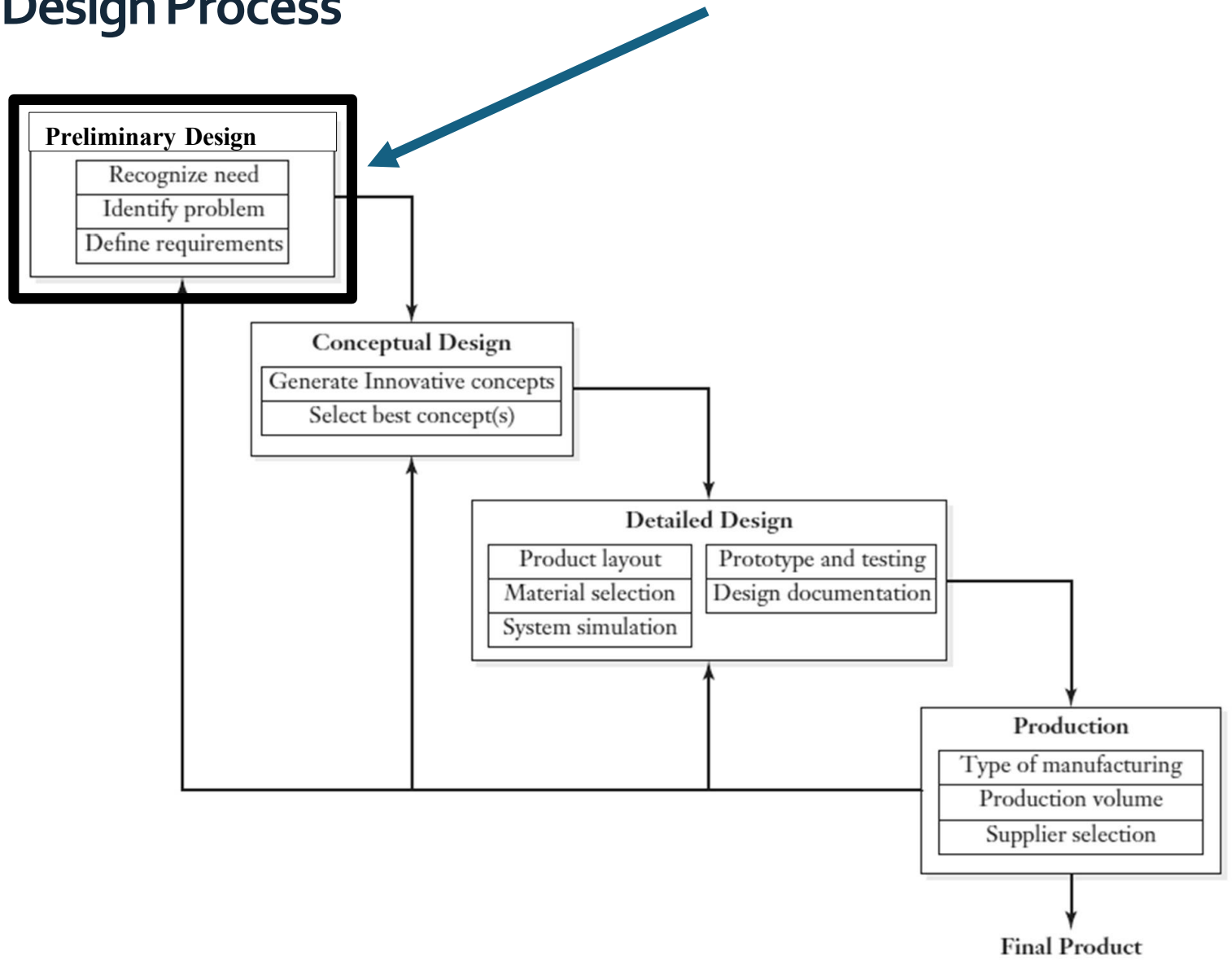
Design Tombola Keywords

- Functional (meets requirements/solves problem/works)
- Long-life (until it completes its function)
- Reliable
- Producable
- Durable
- Cost-effective
- Easy to operate
- Easy to maintain
- Environmentally-friendly

Design Tombola Other Keywords

- Innovative
- Thorough
- Recyclable
- Profitable
- ~~High quality~~ (can be defined in terms of other traits)

The Design Process



Flowchart of the prototypical mechanical design process.

Recognize Need

For Example: At certain times, light streams in the windows of a classroom and makes it hard to see the projected image.



www.greenlearningandliving.wordpress.com

- The need may be given to you by your professor/boss/customer
- You should also do critical observation of the situation when developing the problem definition
- There may be additional constraints you should observe, for example, you cannot change the window location, you cannot change the lecture hour, etc…

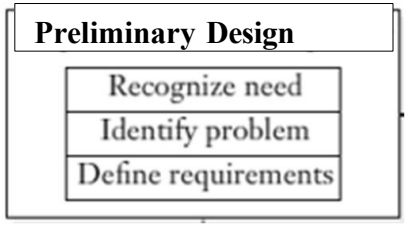
Problem Definition

- Given the need and your observations, there may be several ways of defining the problem
- For example, you may define the problem as follows:
  - Find means to increase the readability of the projected image without changing the window location or the lecture hour.
- Note that, depending on the observations and constraints, the problem definition may be very different.
- The important idea is that the problem definition does not point to a particular solution.
- For example, this is not suitable: Improve the projected image by using a higher quality projector and by covering the window with curtains

Problem Definition

- Problem definition should not repeat the need word by word.
- It should be written by taking into consideration the given need AND any other information that is available, or that you may observe independently.

Define Requirements



- Recognize need Define Problem
- Develop a comprehensive set of system requirements considering the following issues:
  - Functional performance – what the product must accomplish
  - Environmental impact – during production, use, and retirement
  - Manufacturing – resource and material limitations
  - Economic issues – budget, cost, price, profit
  - Ergonomic concerns – human factors, aesthetics, ease of use
  - Global issues – international markets, needs, and opportunities
  - Life cycle issues – use, maintenance, planned obsolescence
  - Social factors – civic, urban, cultural issues
- These requirements represent the constraints that the design must eventually satisfy.

Define Requirements and Design Specifications

- Requirements are full sentences clarifying what your design should be to do.
- For example, as a requirement, "*brightness"* is not acceptable, but "*The image should be readable by all the students "* is.
- The requirements list is then used to develop design specifications
- Design Specifications are measurable quantities and clearly and specifically named. For example "*distance"* is not acceptable, but "*the distance of the seats from the screen"* is. There should be a corresponding limiting value.

Define Requirements and Design Specifications

- While requirements are verbal statements, design specifications are numerical (measurable)

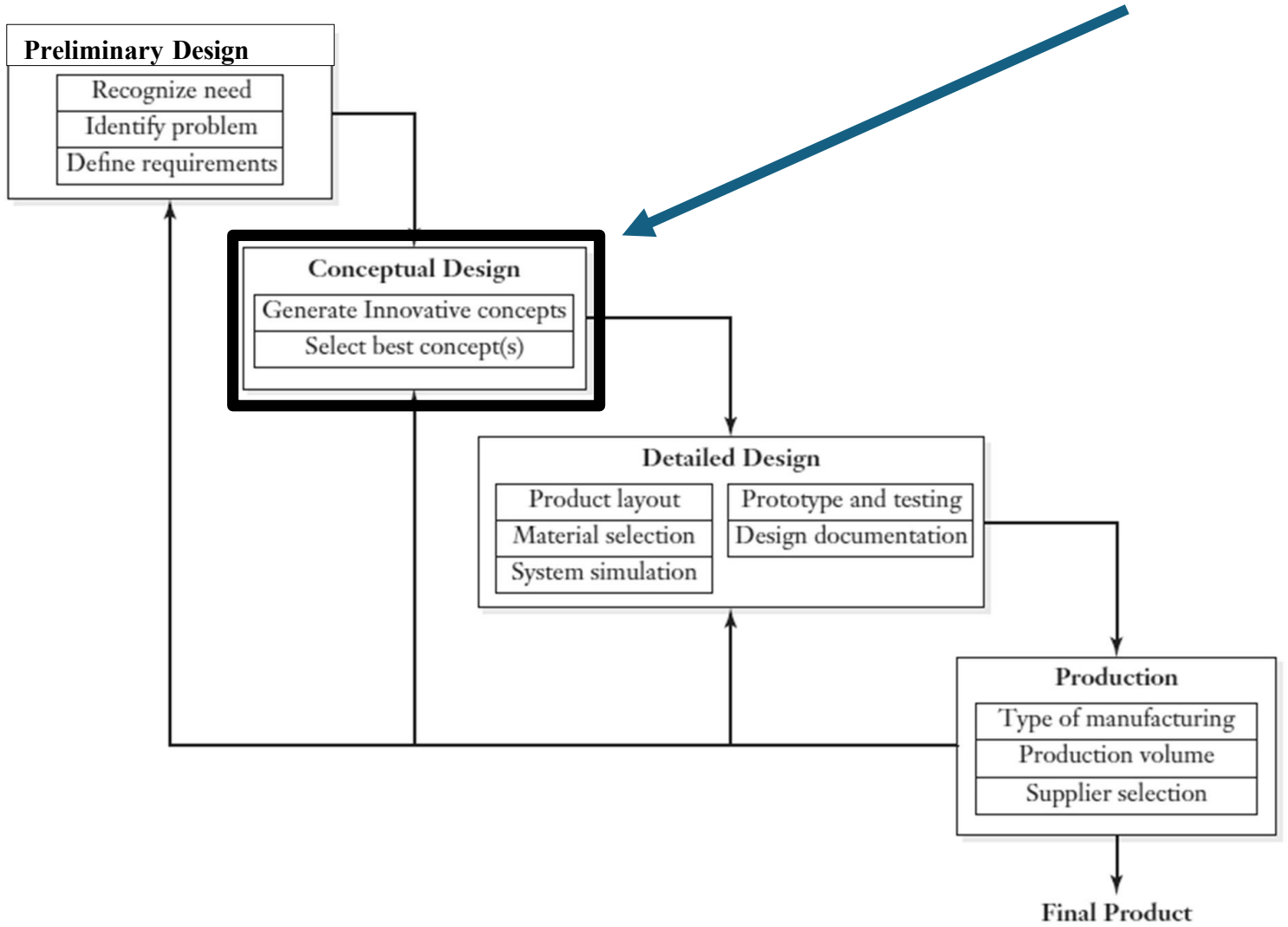
  Ex. Must be light weight → Must be less than 200 g

  (requirement) (design criterion)
- The requirements and the design specifications should match, one-to-one.
- Requirements and design specifications should NOT include anything specific to a possible solution to the problem; such as materials, shape/geometry, or design detail.
- You should write requirements and design specifications that can be satisfied by many possible and different solutions.

Design Process Step 1: Preliminary Design Summary

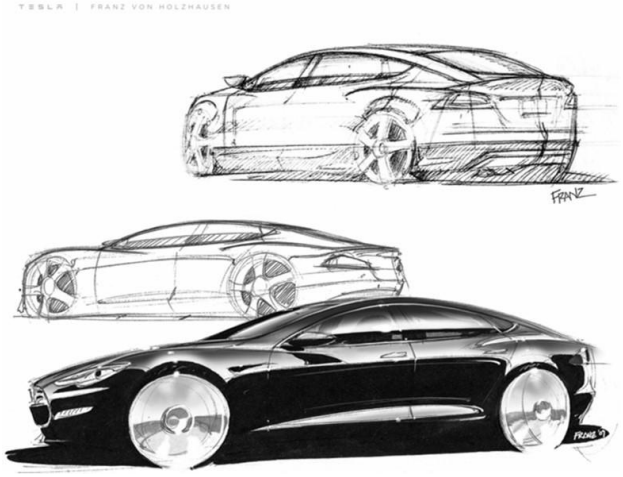
- **Recognize need** from what is given or what you can observe
- **Define problem** based on the need, expectations (of professor/boss/customer), any other observations
- **Define requirements and design specifications** based on what your solution must eventually accomplish

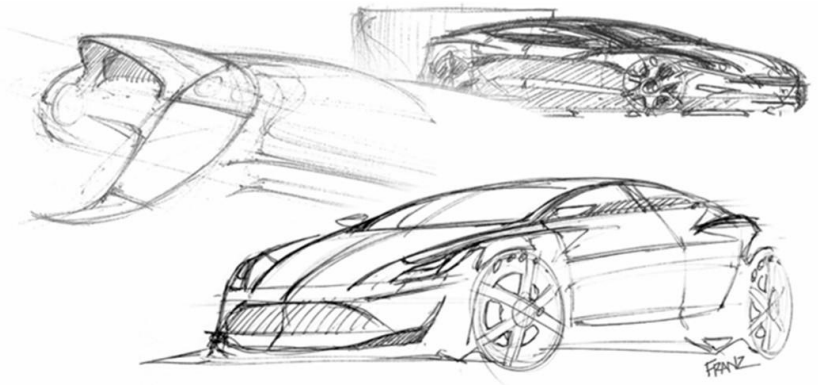
The Design Process



Generating Concepts

- Creativity is a critical part of an engineer's job, as product design requires engineers who are part rational scientists and part innovative artists.
- Engineers work collaboratively and creatively to generate a wide range of potential solutions to the problem at hand, and then select the most promising solution(s) to develop.

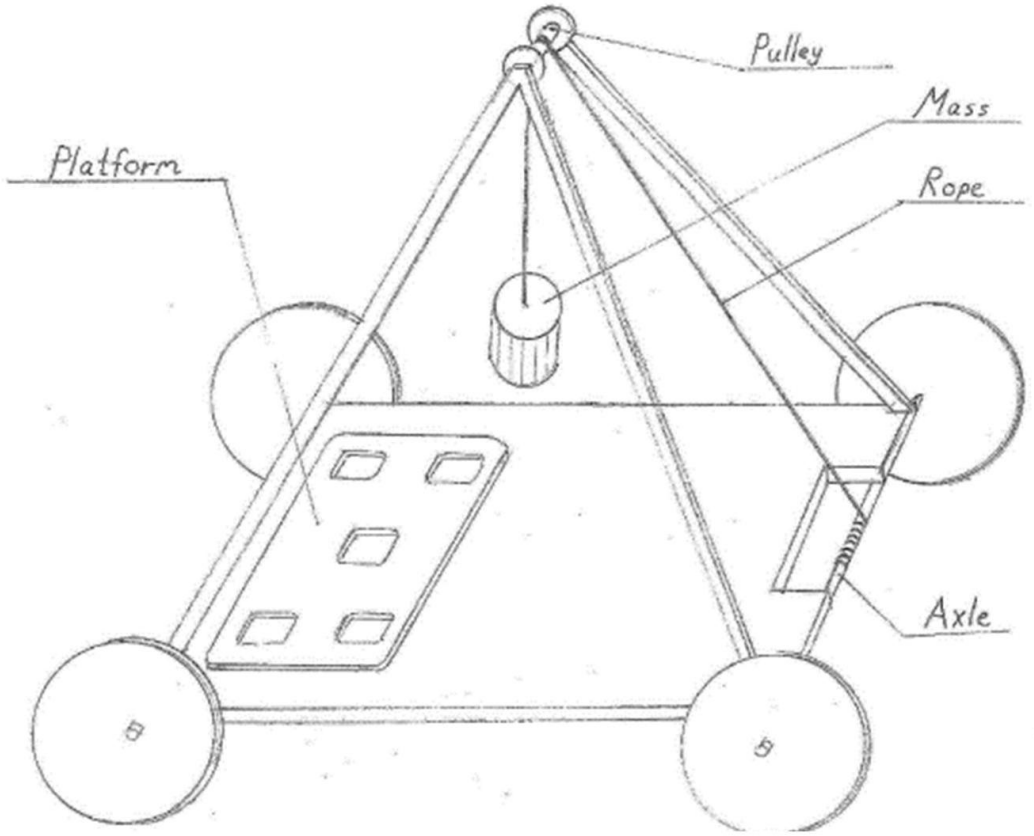




Conceptual Design

- At this stage, sketches are drawn for possible solutions and working principles, or how the concept works and solves the problem, are clearly explained, by specifically addressing the requirements and the design criteria.
- To be understandable, you must indicate parts on your concept sketches, name them, and refer to them in the description.

Conceptual Design Example



Description of Concept:

1. The vehicle has a fixed pulley system on top of it. The rope on the pulley is tied to a mass at one end and wounded around the axle of the vehicle at the other end.
2. As the mass moves down due to gravity, it pulls the rope.
3. Pulled rope causes some torque on the axle and makes the axle turn.
4. To sum up, potential energy of the mass is transformed into kinetic energy of the vehicle.

Conceptual Design

- Design criteria, developed in the preliminary design stage, are used to eliminate infeasible or inferior designs and identify the concepts with the most potential to satisfy the requirements well.
- Eventually, the most promising design concept is selected, using tools such as a Pugh Chart.

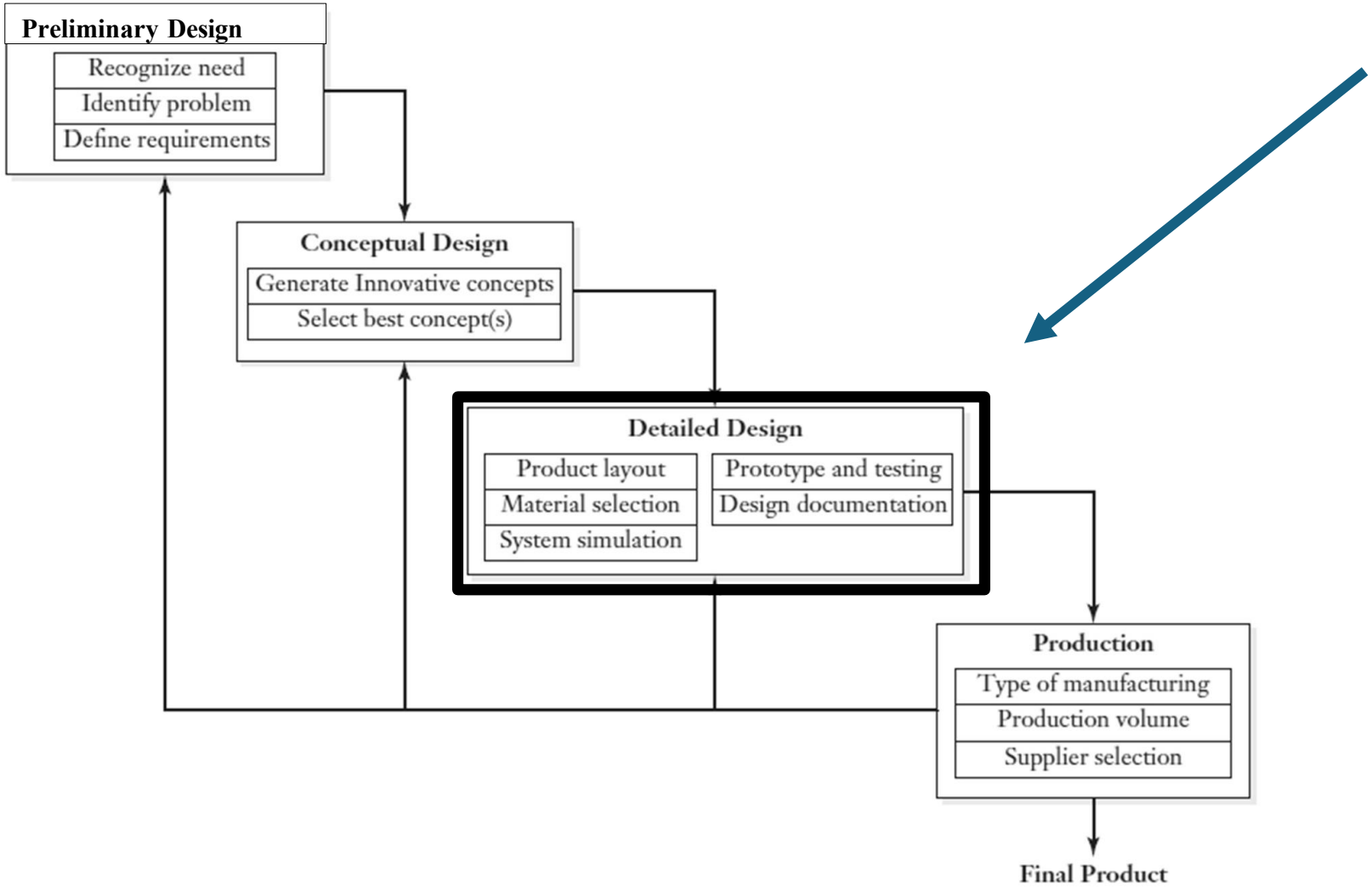
Pugh Chart

- The concepts are compared for several design criteria.
- Each design criteria has a weight (points).
- Among the design concepts, one is arbitrarily selected as the datum (baseline), which must be an "average and acceptable" solution.
- Other concepts are rated relative to this datum, as
  - + (better rating than the datum, add the criterion points)
  - – (worse rating than the datum, subtract the points)
  - S or 0 (same rating as the datum, zero points)
- The concept with the highest points is selected.
- If concepts end up having all higher or all lower points than the datum, select another datum (it was wrong)

Design Process Step 2: Conceptual Design Summary

- **Generate innovative concepts** considering the problem definition and requirements, sketch them, and explain their working principles
- **Select the best concept** using the design criteria and a tool such as Pugh Chart

The Design Process



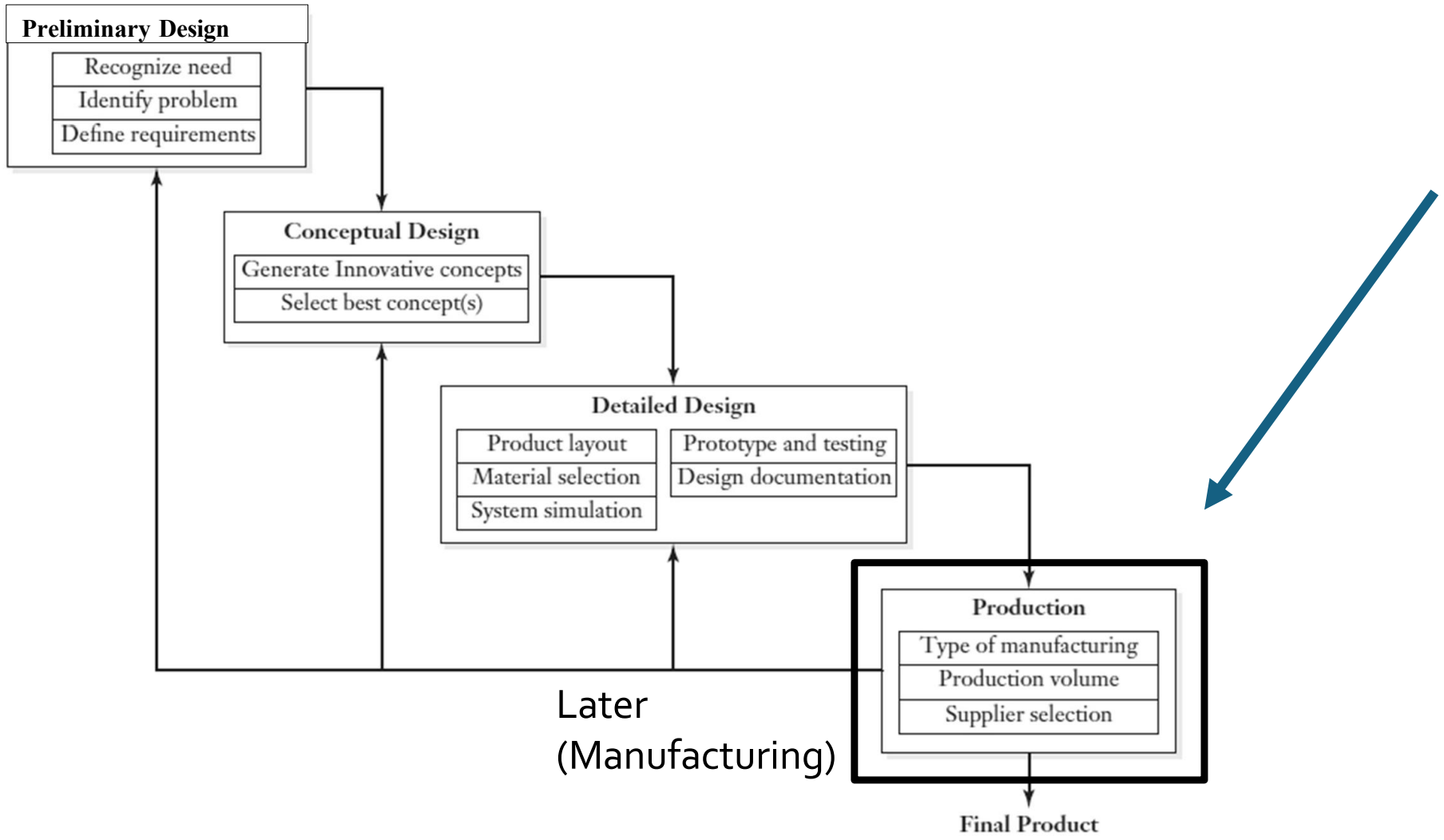
Detailed Design

- In order to build the product, many design and manufacturing details must be determined.
- Detailed design of the product involves:
  - Developing product layout and configuration
  - Selecting materials for each component
  - Optimizing the final geometry, including appropriate tolerances
  - Developing completed digital models of all components and assemblies
  - Simulating the system using digital and mathematical models
  - Prototyping and testing critical components and modules
  - Developing production plans
- The ME curriculum has many courses that will help you accomplish these steps

Some Guiding Principles of Detailed Design

- A simple design is better than a complex one, because fewer things can go wrong.
- Iteration enables to turn hardware that works into hardware that works well. It is the process of making repeated changes and modifications to a design to improve and perfect it.
- No matter how impressive the technology may be, if it is difficult to operate, customers will not like it. Design must be usable.
- Design process must be properly documented.

The Design Process



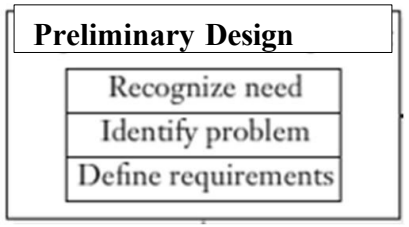
ME 101 Project Example: Self-Energized Vehicle

The following 12 slides are from past ME 110 Student Projects

The Need: A vehicle that travels over a flat surface, without batteries or electricity.

Your project will be evaluated based on the following.

1. Your vehicle must travel the longest distance divided by the total (initial) weight of the vehicle
2. Your vehicle must not use batteries or electricity – your project will not be evaluated if this is violated.
3. Your team can only release the stationary vehicle when the test begins. Once it starts moving, you may not interfere.



Step 1: Preliminary Design

The given need, project evaluation criteria, and any other factors are carefully studied to come up with a suitable Problem Definition.

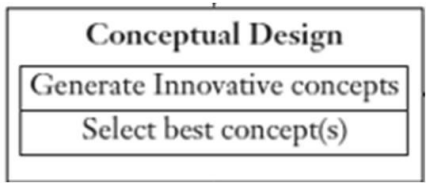
For Example: A light-weight vehicle that travels without batteries or electricity but using other forms of energy, and has the maximum ratio of covered distance divided by the vehicle's weight is to be designed.

Project Requirements List Example

- The weight of the materials that will be used to construct the vehicle should be as small as possible
- The vehicle must be able to travel a long distance to meet the problem's requirement.
- The vehicle should travel straight forward, without rotating or turning sideways.
- The components should be well connected and constructed.The durability should be considered in case of any accidental crash.
- The vehicle should be easily set up, able to travel repeatedly.
- Materials reducing the friction effects should be used.

Design Specifications Example

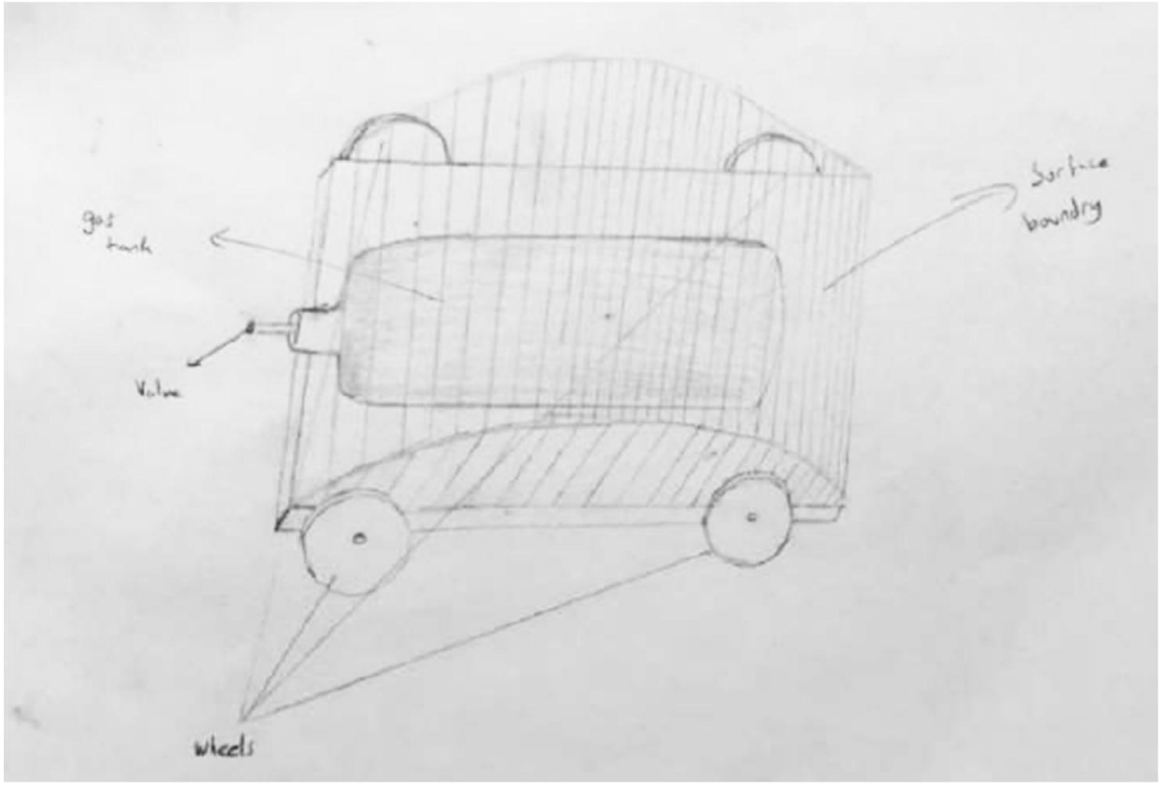
Specifications	Value
Mass of the vehicle	< 200 g
Distance travelled	> 4 m
Travel straight	Max deviation < 1 m
Durability	Minimum test number > 20
Preparation time	< 15 s
Reduce friction	Travel without skidding Yes/No



Concept Generation

- ME 110 Instructors' Expectations: Generate three full concepts for the vehicle and prepare a detailed hand-sketch for each concept
- Concepts should be clearly sketched – computer drawing is not necessary at this point
- All working parts should be understandable
- Each concept (its working mechanism) should be clearly explained

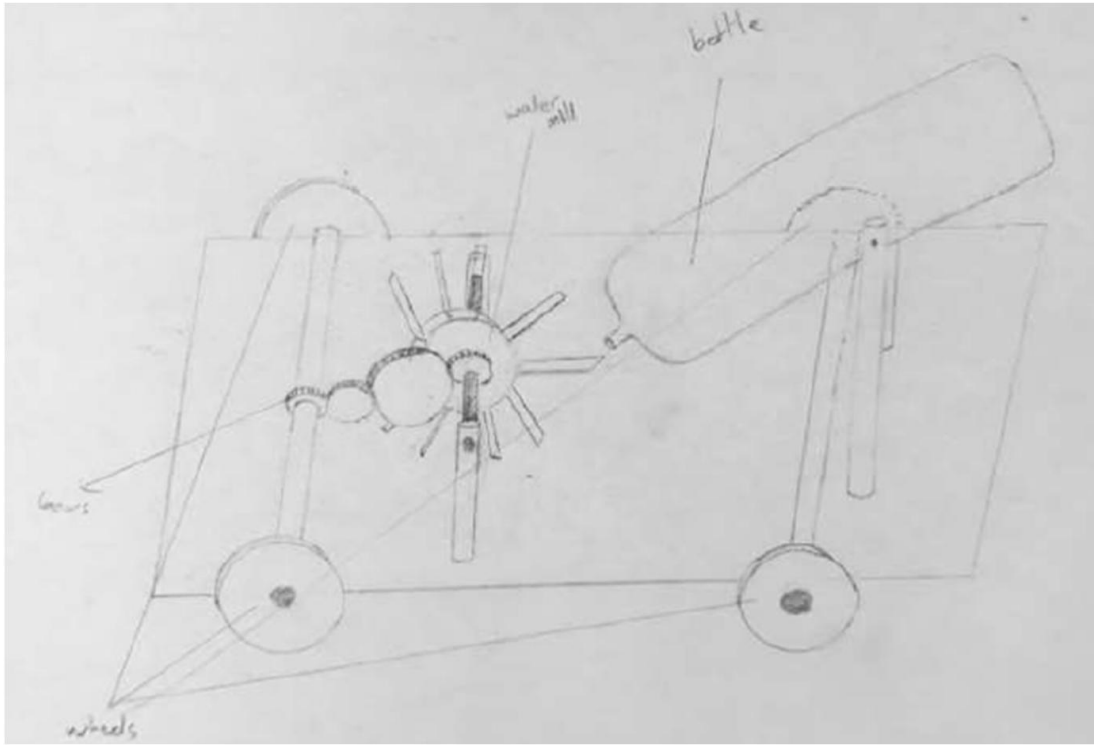
Concept Generation: Concept 1



Pressurized Air Tank Concept

1. The car consists of a compressed gas tube.
2. The car is curved on its top to decrease the air resistance.
3. The gas has high internal potential energy due to compression.
4. The car has a force on it due to change in mass, which causes change in momentum.

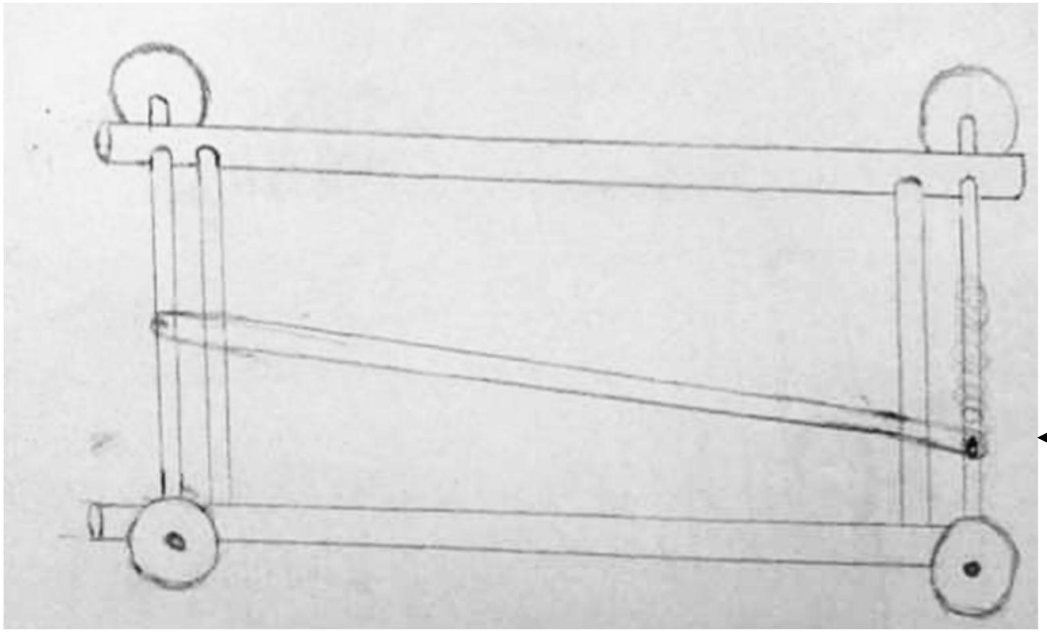
Concept Generation: Concept 2



Water-Propeller Concept

1. The car consist of a propeller and a raised water bottle.
2. The water has potential energy and it comes out of the nozzle with some speed and turns the propeller which turns the axle and runs the car.
3. Since water has high density, the car is heavy and it spills water around.

Concept Generation: Concept 3



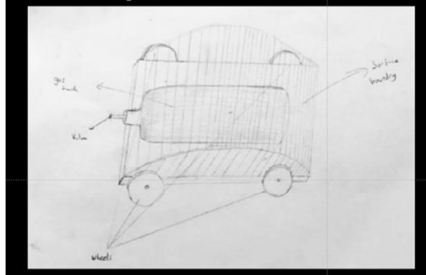
Rubber Band Concept

They should have indicated the parts on the sketch!

1. The car consists of a rubber band which can be wounded over the rear axle of the car.
2. The wounded rubber band creates a torque on the rear axle which makes the car run.

It's Your Turn

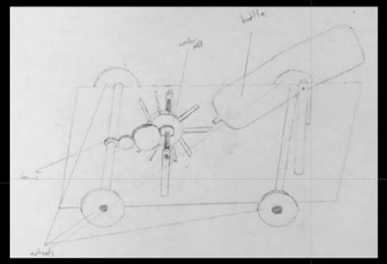
Concept Generation: Concept 1



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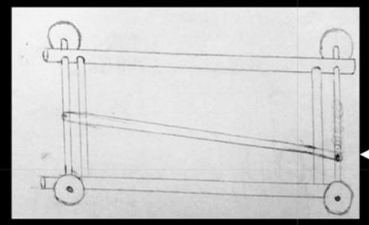
Concept Generation: Concept 2



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Concept Generation: Concept 3



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	Importance	Full Concept 1	Full Concept 2	Full Concept 3
Mass		?	DATUM	
Travel Straight	4			
Durability				
Preparation Time	2			
Distance Travelled	5			
Reduce Friction	3			
Cost	3			
TOTAL				



Compare concept with respect to datum

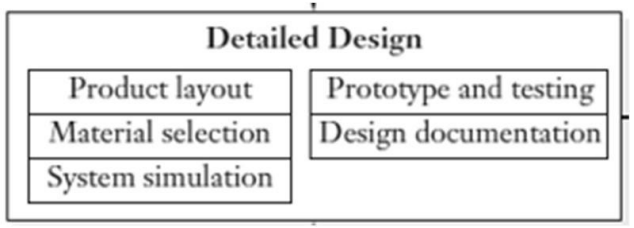
A) Better (+) **B)** Same (S) **C)** Worse (-)

Sample Selection (Pugh Chart)

- The Group selected the Concept 1 (Pressurized Air Tank) as their datum and rated the other concepts with respect to it using a Pugh Chart.
- The importance (weights) for evaluation criteria are selected subjectively.

	Importance	Full Concept 1	Full Concept 2	Full Concept 3
Mass	5	DATUM	S	+
Travel Straight	4		-	-
Durability	3		-	+
Preparation Time	2		-	+
Distance Travelled	5		-	-
Reduce Friction	3		S	S
Cost	3		+	+
TOTAL			-11	4

Full Concept 3 is selected. Detailed design should be done for this concept.



Detailed Design

- Putting together a complete engineering description of the product.
- Drawings with all necessary information for manufacturing.
- Often includes building pre-production versions of the product – prototypes

- Requires having taken many engineering courses.